

Error analysis

Introduction

In this document, we will look at the mistakes our models we making during development and try to correct them by making informed decisions driven by our findings. We first work with 3 separate models, because at the early stage, we still haven't decided on one that can be considered more suitable than the others. However, as we go further down this path of analysis, we should come to one model that is the best and improved.

Models

DistilBERT base uncased finetuned SST-2

This model has a main use for topic classification. Its intended use is not really made for emotion classification, but it is not necessarily mentioned in the documentation that it can't be used for this task. There are some biases in the model, but they are mostly related to biased predictions when it comes to underrepresented populations. In theory, this is not connected to sentiment.

BERT base model (uncased)

This model has been pretrained on a large corpus of English data with self-supervision. Its two main objectives are masked language modeling (MLM) and next sentence prediction (NSP). The model is supposed to be fine-tuned on tasks that include whole sentences to make decisions. This could potentially mean that it can be fine-tuned to predict the emotion of the sentence. However, it is not its intended use for this task.

DistilRoBERTa base

This model is a distilled version of the RoBERTa-base model. This model is also used for MLM. However, it can be fine-tuned for tasks that use the whole sentence to make decisions. We think that this can be used for our task. It introduces some biases, but these are mostly when it comes to the MLM task.

Timeline

These three models all achieved great results when evaluated on the evaluation set, which comprised 20% unseen data from the dataset we also used for training. The best performing one: DistilRoBERTa base yielded a weighted F1 score of 90%. However, when we tested the three models against a completely new dataset, which in our case was the show data annotated by the company's pipeline, the performance dropped drastically. Whereas the

models all had scores in the neighborhood of 90% on the evaluation set, their results against the pipeline's annotations were substantially lower: **DistilBERT base** achieved an **F1 score of 0.387** with an **accuracy of 0.481**, **BERT base** reached an **F1 score of 0.341** and **accuracy of 0.456**, and **DistilRoBERTa base** obtained an **F1 score of 0.435** with an **accuracy of 0.482**.

We attributed these results to the models' learning insignificant patterns in the training data instead of focusing on emotion-filled words or parts of sentences. We believe the training dataset was not aligned with the show data, and that is why the model did so poorly when evaluated against the pipeline. We looked at the training dataset and identified a potential mismatch between the training dataset and the test dataset. The training dataset mostly contained sentences from the "**Sentiment and Emotion Analysis Dataset**". These sentences have the shape "I am {...}", "I feel {...}", "I like {...}", "I hate {...}". When sampling the dataset, the training and evaluation datasets use this dataset only for the emotions with more samples, and only use the other two datasets that are part of the dataset we created for this task when the emotion is undersampled in the "**Sentiment and Emotion Analysis Dataset**". We believe that this makes the model learn to do well when the sentences have this "I {...}" shape, but they don't generalize well when the sentences have different shapes.

We tried to mitigate this by first removing all the sentences from the "**Sentiment and Emotion Analysis Dataset**" from our training and evaluation set. This way, the model would see sentences with different shapes and sizes, and we believed this would increase its generalization capabilities and would make it more robust to different test datasets. This change also diminished the number of samples we had per class, leading to the lowest represented class only having 4.8k rows of data.

We also identified another issue. When we initially fine-tuned the three transformer models, we didn't freeze any of their layers, meaning that all the parameters in the model got changed when we retrained them. We believed this could also be a reason for the models' bad performance on the show data and why they couldn't generalize across datasets. To overcome this, we wrote code that froze all the parameters in the models' bodies and heads, and we had additional flags that could unfreeze the classifier and some transformer blocks from the models' bodies.

After we addressed the issues we found with how we fine-tuned the models as described above, we began retraining the models using the new training dataset and the ability to select how many layers we want to retrain in the models' bodies. We experimented with different parameters to see which combination yielded the best results. We identified a learning rate of $1e-5$, a batch size of 32, a weight decay between 0.01 and 0.05, and the number of unfrozen transformer layers between 1 and 3 as being good starting points for all models. We observed that the best-performing models were the DistilBERT and the DistilRoBERTa models. BERT base models yielded the worst performance out of everything we tested, so we decided to exclude them from future tests. The performance on the show data also increased slightly with the DistilBERT and the DistilRoBERTa models, achieving performances between 51-55% depending on the exact parameter configurations we used. We also observed that the number of unfrozen layers was the parameter impacting the performance the most.

The next step in improving our models' performance on the show data involved expanding the training dataset. To do this, we incorporated the GoEmotions dataset into our existing data. We aligned the emotion labels from GoEmotions with our existing emotion mapping and then merged the two datasets into a single, larger training set. This increased the number of samples for all classes, allowing us to sample more examples for each class, which could potentially make our models better.

We began retraining our models again with this newly created, larger dataset. We again experimented with different model parameters until we found a configuration that worked best. This time, we also experimented with using the entire dataset versus sampling it so all classes had similar numbers of rows. Our observation was that in almost all cases, balanced data was better than feeding the model the entire dataset, even though that meant less data. We also observed a behavior that we didn't expect. We noticed that across multiple runs, the DistilBERT models were performing better than the DistilRoBERTa models. With this in mind, and also the fact that DistilRoBERTa models have more parameters, which makes them more computationally and time-intensive, we decided to exclude DistilRoBERTa models from future iterations. We achieved a satisfactory performance with the DistilBERT model trained on a balanced dataset of 12K samples per class, a learning rate of $1e-5$, a batch size of 32, a weight decay of 0.0, and the number of unfrozen transformer layers being 2. This model achieved scores much higher than what we had before, reaching an accuracy of 64% and a weighted F1 score of 61%. We were satisfied enough with this performance to conduct error analysis and explainable AI techniques on this model in the hope that these explainability techniques would allow us to identify additional flows in the model that we would be able to act upon.

Error Analysis

We began conducting error analysis on the best-performing model: the DistilBERT model, fine-tuned on our specific task. After reviewing the results, we noticed that the company's pipeline produced several misclassifications. Since our models were meant to be evaluated against their outputs, we decided to review all the results and correct any mistakes to ensure a more accurate error analysis process. Throughout the error analysis, we used our manually assigned classification labels as the ground truth for evaluating and comparing the performance of our models.

Metrics and Classification Report

The first step in our error analysis was looking at different metrics in the classification report. As mentioned previously, the model had an accuracy of 64.1% and a weighted F1 score of 61.7%. We also looked at the classification report to get a better understanding of the errors made by the model.

--- Classification Report ---			
	precision	recall	f1-score
neutral	0.69	0.87	0.77
anger	0.66	0.31	0.42
disgust	0.61	0.25	0.35
fear	0.61	0.25	0.35
happiness	0.70	0.53	0.60
sadness	0.43	0.36	0.39
surprise	0.46	0.73	0.56
accuracy			0.64
macro avg	0.59	0.47	0.49
weighted avg	0.64	0.64	0.62

The classification report provides a detailed breakdown of the model's performance across each emotion class. It includes precision, recall, and F1-score, which together help identify where the model performs well and where it struggles.

Precision indicates how many of the predicted samples for a class were correct. High precision means the model makes few false positives. For example, the model performs best in predicting *happiness* (0.70) and *neutral* (0.69), suggesting that when it predicts these emotions, it is often correct. In contrast, *surprise* (0.46) and *sadness* (0.43) have low precision, indicating frequent misclassifications.

Recall measures how many actual samples of a class were correctly identified by the model. Here, *neutral* has a high recall (0.87), showing that most neutral samples were correctly detected. However, *disgust* and *fear* have notably low recall (0.25 each), implying that the model misses many true instances of these emotions.

F1-score combines precision and recall into a single measure, offering a balanced view of performance. The model achieves its best F1-scores for *neutral* (0.77) and *happiness* (0.60), while *disgust* and *fear* perform poorly (0.35 each). This imbalance suggests that the model is biased towards recognizing more prevalent or easily distinguishable emotions, such as neutral expressions, while struggling with subtler or less represented classes.

The **macro average** treats all classes equally, showing a relatively low F1-score of 0.49, which highlights that overall performance varies significantly between emotions. The **weighted average** accounts for class distribution, resulting in a higher F1-score of 0.62, suggesting that the model performs reasonably well on the more frequent classes.

Overall, the classification report demonstrates that while the model performs reasonably well on dominant emotions such as neutral and happiness, it performs poorly on less frequent or more subtle emotions like fear, disgust, and sadness. This significant disparity in performance between dominant and underrepresented emotions indicates that, despite being trained on a balanced dataset, the model struggles to recognize all emotions with equal accuracy.

Misclassified Examples

The next step in our error analysis process was to look at some examples of misclassifications made by the model. We took the first 20 such examples and printed them out to inspect them.

Upon manual inspection, the misclassifications can be divided into three categories: **true model errors**, **potentially mislabeled data**, and **ambiguous or context-dependent sentences**.

```
--- Misclassified examples for distilbert-base (Sample, First 20) ---
```

Sentence	True_Emotion	Pred_Emotion
this suburb is home to a neighborhood Italian bistro	happiness	neutral
but I didn't have the money or the collateral,	sadness	neutral
finance the restaurant and put their houses up as collateral	fear	neutral
to help Tatiana fulfill her dream.	happiness	neutral
My mom and I don't know about the restaurant business,	neutral	sadness
Are we going to have enough bread for tonight?	fear	neutral
But it didn't pan out the way I wanted it to.	sadness	neutral
and have maybe 4 tables at the most.	sadness	neutral
I only had one.	sadness	neutral
Who's going to bring the people in?	fear	neutral
The problem with the restaurant is Tatiana.	anger	neutral
The fact that she feels like	neutral	fear
No, they're crazy.	anger	neutral
and he's pretty awful.	anger	disgust
It's a little chewier, actually.	disgust	neutral
Do we have any tables that are happy?	fear	neutral
She's very disrespectful.	anger	disgust
Tatiana doesn't know how to handle being an owner.	sadness	neutral
Don't drink so much, okay?	fear	neutral
My mom and my sister already resent me	anger	sadness

1. True model errors

These are sentences where the true emotion labels appear accurate, but the model failed to predict them correctly:

- “*but I didn't have the money or the collateral,*” → True: **sadness**, Pred: **neutral**
→ Clearly expresses regret, indicating sadness.
- “*The problem with the restaurant is Tatiana.*” → True: **anger**, Pred: **neutral**
→ This is a direct statement of frustration; the model underestimates negative emotion.
- “*She's very disrespectful.*” → True: **anger**, Pred: **disgust**
→ The emotion aligns more with anger than disgust; a genuine classification error.
- “*My mom and my sister already resent me.*” → True: **anger**, Pred: **sadness**
→ While resentment involves sadness, the tone conveys bitterness or frustration, making anger the more fitting label.

2. Potentially mislabeled true emotions

In these cases, the given *true* emotion label appears questionable or debatable:

- *“to help Tatiana fulfill her dream.”* → True: **happiness**, Pred: **neutral**
→ The sentence itself lacks an explicit emotional signal; the label might be overstated.
 - *“and he’s pretty awful.”* → True: **anger**, Pred: **disgust**
→ The wording better fits disgust than anger, so the model’s output may actually be more appropriate.
 - *“It’s a little chewier, actually.”* → True: **disgust**, Pred: **neutral**
→ This is likely mislabeled; no clear disgust is expressed in the sentence.
-

3. Ambiguous or context-dependent sentences

These sentences could validly belong to more than one emotion class, or require additional context to interpret:

- *“Are we going to have enough bread for tonight?”* → True: **fear**, Pred: **neutral**
→ Without context (e.g., financial stress or anxiety), it could be neutral.
 - *“I only had one.”* → True: **sadness**, Pred: **neutral**
→ The sentence itself is short and vague, allowing it to have multiple emotions associated with it.
 - *“Who’s going to bring the people in?”* → True: **fear**, Pred: **neutral**
→ The emotion depends on tone or situation; it may indicate concern or simple inquiry.
 - *“Do we have any tables that are happy?”* → True: **fear**, Pred: **neutral**
→ Could be rhetorical, sarcastic, or literal—emotion unclear without dialogue context.
 - *“Don’t drink so much, okay?”* → True: **fear**, Pred: **neutral**
→ Could be fear, care, or simple advice depending on tone.
 - *“No, they’re crazy.”* → True: **anger**, Pred: **neutral**
→ Ambiguous—could express irritation, disbelief, or humor depending on delivery.
-

Summary

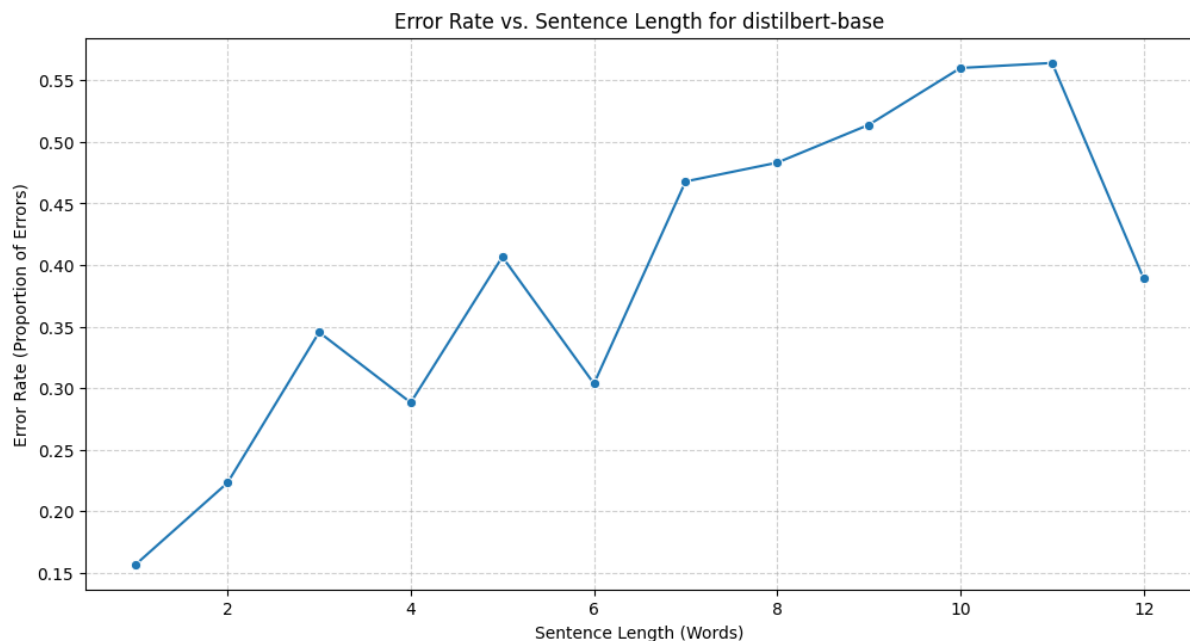
Out of the 20 examples:

- **4-5 sentences** appear to be genuine model misclassifications.
- **3-4 sentences** likely have incorrect or debatable emotion labels.
- **The remainder** are ambiguous or context-sensitive, where emotional inference requires additional contextual cues.

This indicates that not all “errors” arise from model weaknesses; some stem from label noise and the inherent subjectivity of emotion annotation.

Sentence Length VS. Error Rate Analysis

The next step was looking at the length of the sentences and comparing them against the error rate for that sentence length.



As shown in the graph, there is a clear relationship between sentence length and error rate for the *DistilBERT-base* model. Shorter sentences (1–2 words) exhibit relatively low error rates, remaining below 0.25. However, as sentence length increases, the error rate begins to rise, suggesting that the model struggles more with longer or syntactically complex inputs.

Between 3 and 5 words, there is noticeable fluctuation, with error rates ranging from approximately 0.29 to 0.41, indicating inconsistent model performance for medium-length sentences. From 7 to 10 words, the error rate continues to increase, peaking around 0.56, which implies a significant degradation in prediction accuracy for longer sentences.

Interestingly, beyond 11 words, the error rate drops again to around 0.39, possibly due to fewer long sentences in the dataset or because longer contexts occasionally help disambiguate meaning.

Overall, the trend suggests that the *DistilBERT-base* model performs best on shorter sentences and becomes less reliable as sentence length grows, with a pronounced performance decline around the mid-to-long sentence range.

This behavior indicates that, since the model was fine-tuned on generic sentiment classification datasets, it exhibits limited robustness when handling longer sentences containing subtle or ambiguous emotional cues. This highlights the need to develop a model capable of effectively leveraging contextual information from preceding sentences. In emotion classification tasks involving transcribed data from television shows, earlier sentences within the same dialogue or scene often play a crucial role in accurately interpreting and labeling the emotions expressed in later sentences. We strongly believe that incorporating contextual information from prior sentences will enhance the model's overall performance.

Top Misclassification Pairs

In this section, we explore the top 5 pairs of missclassified examples.

```
--- Top Misclassification Pairs (True -> Predicted) ---
True_Emotion Pred_Emotion Count
      anger      neutral    74
    happiness      neutral    61
      sadness      neutral    37
      neutral    happiness    29
      anger      sadness    25
```

We observed that most of the model's misclassifications occur when labeling sentences as neutral. This reveals a limitation in the current approach, where the model tends to default to the neutral label when uncertain about the appropriate emotion. One potential mitigation strategy is to introduce a two-stage model: the first stage determines whether a sentence is neutral or contains an emotion, and the second stage, applied only to emotionally charged sentences, classifies the specific emotion. This separation prevents confusion between the six core emotions and the neutral category, which is not an emotion itself. Among the top five misclassification types, the two least frequent are *neutral* → *happiness* and *anger* → *sadness*, with 29 and 25 occurrences, respectively.

Grave VS. Understandable Error Classification

After identifying the top 5 misclassification types made by our model, we wanted to assess the **gravity** of those mistakes. A mistake was defined as grave if the model misclassified an emotion's fundamental sentiment, specifically:

- Positive-to-Negative: Classifying a positive emotion, such as 'happiness', as a negative emotion ('anger', 'disgust', 'fear', or 'sadness').
- Negative-to-Positive: Classifying a negative emotion as a positive one.

In contrast, **understandable errors** were defined as misclassifications involving the more ambiguous or neutral emotions, such as classifying a positive or negative emotion as '**neutral**' or '**surprise**', or vice versa.

```
--- Error Gravity Analysis ---
Error_Type
Understandable          97.50
Grave (Negative to Positive)  2.27
Grave (Positive to Negative)  0.23
```

The table above reveals that the vast majority of the model's top misclassifications, 97.5%, are **understandable errors**. This indicates that the model is highly effective at distinguishing the

core sentiment of an emotion. When it does make a mistake, it typically confuses the target emotion with one that shares a similar valence (e.g., neutral with happiness, or anger with sadness), rather than flipping the sentiment entirely.

The **grave mistakes** constitute a very small portion of the overall error set, totaling only 2.27% + 0.23% = 2.5%.

- The most common grave error is **Negative to Positive** at 2.27%. This means the model is slightly more likely to confuse a negative emotion (like anger or sadness) for a positive one (like happiness).
- The rarest grave error is **Positive to Negative** at a mere 0.27. This is an excellent result, showing the model seldom misinterprets a clear positive emotion as a negative one.

In summary, the model demonstrates a high degree of reliability in maintaining emotional **valence**, with grave errors being a negligible fraction of its total misclassifications.